

AICO Data Inputs, Outputs, and Measurement Points

A complete reference to every data surface in the AI Capability Observatory

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AICO (AI Capability Observatory) is a **receiver and scoring platform**, not an execution engine: it never runs a model, harness, or task itself. Every number it reports is either (a) a value someone submitted, or (b) a statistic computed from submitted values. This page enumerates every point where data enters the system, every point where a measurement is produced, and every point where a result leaves the system – so any consumer can trace a number on a dashboard all the way back to the raw observation(s) that produced it.

Keep this page current: this is a living reference. Whenever a new input endpoint, computed statistic, or output surface is added to AICO, update this page (and regenerate its PDF companion via `scripts/generate_docs_pdfs.py`) in the same change.

RAW INPUTS	MEASUREMENT ENGINES	OUTPUTS
Benchmarks/Metrics	->	
Observations	-> Hierarchical Bayesian IRT Fit	-> Leaderboard, Model
	(Global->Source->Domain->	Scores, Snapshots,
	Benchmark->Metric)	History

	+-----+	
	Drift / Uncertainty / Forecast / Gap / Attribution / Portfolio / Graph	
	MITRE ATLAS technique tags -> Coverage & per-model Exposure rollup	
Harness Trial	-> Bradley-Terry / Glicko-2 /	-> Harness Leaderboard,
Results	Pareto Frontier / GLMM /	Drift, Frontier,
	Maturity radar	Maturity, Variance
	-----	Decomposition
Synthetic Study	-> Simulation Lab (recovery/	-> Recovery/Coverage
Parameters	coverage validation)	Report (validates the
	-----	measurement engine itself)
Every fit	-> Reproducibility Manifest	-> Governance Audit Trail
publication	(seeds, versions, git commit)	& sealed Manifests

Raw data inputs

Everything AICO measures traces back to one of these submission points. None of these endpoints compute anything themselves – they only validate and persist.

#	Input	Endpoint	Fields	What it represents
1	Benchmark registration	POST /benchmarks	benchmark_id, domain_id, source_type (public/internal), tier (1-4), business_unit; one or more MetricSpec (metric_name, metric_type)	Declares a benchmark exists and which metrics it reports
2	Observation	POST /observations	model_id, benchmark_id, metric_name, value	The atomic unit of measurement – one model’s raw score on one metric of one benchmark
3	Harness trial result	POST /harness/results	Identity (run_id, harness/model/task); outcome (pass/fail, tests passed/total); metrics (elapsed time, tokens, cost, tool calls, files modified, commands executed); provenance (engine, timestamps, sandbox digest)	One agentic-coding-harness trial, already executed by an external system
4	Harness artifact	POST /harness/artifacts	raw bytes (transcript, diff)	Supporting evidence for a trial result – stored, never itself measured
5	MITRE ATLAS tags	PUT /benchmarks/{id}/mitre	technique_ids[]	Human-curated mapping of a benchmark to the adversarial-ML techniques it exercises
6	Model/benchmark visibility	PUT /models/{id}/restrict PUT /benchmarks/{id}/restrict	restricted, allowed_user_ids[]	Access control – governs <i>who sees</i> a score, not the score itself
7	Model hosting record	PUT /models/{id}/hosting	hosting_system (free text)	Informational only – never enters the scoring model

#	Input	Endpoint	Fields	What it represents
8	Model commercial catalog	PUT <code>/models/{id}/catalog</code>	provider, pricing, context window, release date, etc.	Feeds Rocky's recommendation engine, not the IRT fit
9	OIDC identity	POST <code>/auth/callback</code>	verified ID token	Establishes <i>who</i> submitted data (recorded on every audit event)
10	Rocky LLM config	PUT <code>/assistant/config</code>	provider, endpoint, model, API key	Operational configuration for the assistant, not a measurement input
11	Simulation study parameters	POST <code>/simulations/jobs</code>	<code>kind</code> (recovery/coverage), <code>method</code> , ecosystem size, <code>n_replications</code> , <code>hdi_prob</code>	Defines a synthetic dataset whose ground truth is known by construction

Core measurement engine: the hierarchical Bayesian fit

Triggered by POST `/recompute` (synchronous) or POST `/recompute/jobs` (background job); also run internally by demo-data loading. Every read endpoint below serves the output of the **most recently published** fit – nothing is measured live on a read.

What is measured, layer by layer (each layer's output is a `CapabilityResult`: `mean`, `median`, `hdi_low`/`hdi_high` (95% credible interval), plus `rank_mean`/`rank_hdi_low`/`rank_hdi_high`):

Layer	Measures	Derived from
Metric (input)	One observed value, typed per <code>MetricType</code> : Bernoulli (pass/fail), Binomial (k of n), Normal (continuous score), Beta (proportion), Ordered Logistic (categorical rating)	Raw Observation
Benchmark	A model's latent ability on one benchmark (2PL/3PL IRT: discrimination, difficulty, optional guessing)	All observations against that benchmark
Domain	A model's ability across all benchmarks in one domain	Benchmark-layer latents, LKJ-correlated residuals
Source	Public vs. Internal capability	Domain-layer latents, split by <code>source_type</code>
Global	Overall Internal Capability Index (ICI)	Source-layer latents

Convergence diagnostics (R-hat/ESS/divergences for NUTS; ELBO-plateau check for VI) are measured and stored alongside every fit as `convergence_passed`/`convergence_failures`.

Derived measurements (computed from fit history, not raw data)

These never touch raw **Observations** directly – they operate on the sequence of already-published **CapabilityResults**.

Measurement	Endpoint	What it computes
Leaderboard	GET /leaderboard	Global ICI, ranked, with rank uncertainty
Score history	GET /snapshots, GET /history/{model_id}	Every published score for a model, across every fit ever run
Drift	GET /drift	Consecutive-fit delta per model; classified stable/watch/drifted against a tolerance
Forecast	GET /forecasts/{model_id}	State-space projection of future capability, with probability-at-or-above a threshold
Generalization gap	GET /generalization-gap	Public ICI minus Internal ICI, computed draw-wise (not from summarized means) so shared-latent uncertainty is preserved
Value/rank uncertainty	GET /uncertainty	Width of each model's credible interval and rank interval, at any layer
Attribution	POST /attribution	Shapley decomposition of a score change between two fits into per-unit (domain/benchmark) contributions
Portfolio health	GET /portfolio	Fisher information, redundancy, and coverage of the benchmark suite itself – measures the <i>test</i> , not the models
Capability graph	GET /graph	Centrality, community structure, and bottlenecks in the domain/benchmark/model relationship graph
MITRE coverage	GET /mitre/coverage	Which registered benchmarks exercise each ATLAS technique
MITRE exposure	GET /mitre/exposure/{model_id}	A model's score, rolled up per ATLAS technique it has been evaluated against (weighted by benchmark score)
Executive decisions	POST /decisions, POST /decisions/what-if	Procurement/ROI/budget recommendations, combining published scores with submitted cost/business constraints

Harness comparison track (a parallel, independent measurement pipeline)

Harness trials are never IRT-scored (a harness is not a clean test-taker – see the “Why not IRT” rationale in [AI harness comparison](#)). Four of the five measurements below are computed fresh on every read; the fifth is a deliberate, admin-triggered exception.

Measurement	Endpoint	Computed on read?	What it measures
Bradley-Terry leaderboard	GET /harness/leaderboard	Yes	Pairwise-comparison strength per harness, with bootstrap 95% CI

Measurement	Endpoint	Computed on read?	What it measures
Head-to-head comparisons	GET /harness/comparisons	Yes	Every (model, task) result the leaderboard was derived from
Cost/accuracy frontier	GET /harness/frontier	Yes	Pareto-optimal (cost, pass-rate) points per (harness, model)
Continuous monitoring	GET /harness/drift	Yes	Glicko-2 rating + rating deviation per (harness, version) across submitted <code>run_id</code> periods
Maturity radar	GET /harness/maturity	Yes	Five-axis operational profile per harness (correctness, speed, cost efficiency, tool engagement, edit footprint), min-max normalized within the harnesses compared
Variance decomposition	POST /harness/variance-decomposition/jobs, then GET /harness/variance-decomposition	No – published job	A GLMM’s attribution of outcome variance to harness vs. model vs. task vs. residual, with 95% credible intervals

Meta-measurement: validating the measurement system itself

Measurement	Endpoint	What it measures
Recovery study	POST /simulations/jobs (kind=recovery)	How well the fit recovers <i>known</i> ground-truth parameters (correlation, RMSE, mean bias) from data forward-sampled from the model’s own prior
Coverage study	POST /simulations/jobs (kind=coverage)	Whether the model’s 95% credible intervals actually contain the true value ~95% of the time, across many replications (empirical_coverage, binomial p-value)
Reproducibility manifest	sealed automatically on every POST /recompute	Git commit, package versions, sampler seed/config, environment snapshot – measures <i>process integrity</i> , not model capability
Audit trail	GET /governance/audit	Every mutating action (who, what, when) – measures <i>who changed what</i> , for compliance

See [Verification & Quality Assurance](#) for the full inventory of tests, scanners, and gates that verify AICO itself behaves as intended – a different question from what this page answers.

Output / consumption surfaces

Output	Endpoint(s)	Consumer
Leaderboard, model detail, domain rankings	GET /leaderboard, /models/{id}, /domains	/, /models/{id} pages
Snapshot & score history	GET /snapshots, /history/{id}	/settings/snapshots, trend charts
Drift, uncertainty, forecasts, gap	GET /drift, /uncertainty, /forecasts/{id}, /generalization-gap	/trends, /stability, /gap, /uncertainty pages
Portfolio & graph	GET /portfolio, /graph	/portfolio, /graph pages
MITRE coverage/exposure	GET /mitre/*	/mitre page
Harness leader-board/frontier/drift/maturity/variance	GET /harness/*	/harness page
Executive decision support	POST /decisions	/decisions page
Governance/audit exports	GET /governance/*	/settings/governance page, compliance review
Natural-language answers	POST /assistant/chat/stream, /assistant/query	Rocky widget – synthesizes all of the above via tool calls, never a separate measurement source
Prometheus metrics	GET /metrics	Grafana – system health (fit duration, convergence, request latency, chat/tool metrics), a fundamentally different measurement class from AI capability itself

Summary: the three measurement classes

1. **Capability measurement** – what an AI model or harness can do (the core fit and the harness comparison track). Always traceable to a submitted **Observation** or **HarnessTrialResult**.
2. **Process measurement** – whether the measurement system itself is trustworthy (the meta-measurement section above, and [Verification & Quality Assurance](#) in full). Recovery/coverage studies, reproducibility manifests, and every automated test/scan/gate.
3. **Operational measurement** – whether the platform is healthy (Prometheus/OTel). Not a statement about any AI system’s quality.

Every dashboard number in AICO is one of these three; none are computed from anything AICO itself executed.

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